

Airbag ECU Portfolio
HS Harz — Functional Safety Engineering

AIRBAG ELECTRONIC CONTROL UNIT

Measurement and Processing of Sensor Signals to Control Airbag Operation

INTEGRATION TEST PLAN (ITP)

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1 Purpose and Scope

This Integration Test Plan (ITP) defines the test strategy, test environment, entry and exit criteria, and full requirement traceability for verifying the interfaces and interactions between firmware modules and between hardware and software in the Airbag ECU system. It covers integration-level and HW/SW bring-up testing only; unit testing and validation testing are addressed in separate documents.

The plan covers the following integration test activities mandated by ISO 26262-6:2018 for ASIL D software:

- Integration Testing — inter-module and HW/SW interaction verification (IT-01 to IT-15)
- HW/SW Integration Bring-Up — 12-step flash, pin, CAN, and AND gate verification procedure

This plan must be read in conjunction with:

- SADS-AIRBAG-ECU-001 Rev 2.0 — Software Architecture Design Specification and Implementation
- SRS-AIRBAG-SW-001 Rev 2.1 — Software Requirements Specification
- FSRs-AIRBAG-ECU-001 Rev 2.1 — Functional Safety Requirements Specification
- HADS-AIRBAG-ECU-001 Rev 2.1 — Hardware Architecture and Design Specification

NOTE: Eight open compliance gaps (GAP-01 through GAP-08) exist in the current firmware, as documented in SADS-AIRBAG-ECU-001 Rev 2.0 Section 10. These are addressed in Section 9 of this plan with a risk acceptance or closure status for each. Tests that exercise unimplemented features are marked accordingly.

2 Test Levels and Strategy

2.1 Test Level Overview

Test Level	Scope	SADS Reference	Test Case Range
Integration Testing	Module-to-module + HW/SW interaction on fully assembled system	Section 9 (SADS Rev 2.0)	IT-01 to IT-15
HW/SW Integration Bring-Up	12-step flash, pin verification, CAN bus bring-up, AND gate verification	Section 9 (SADS Rev 2.0)	Bring-up Steps 1–12

2.2 Test Approach

All tests are executed on target hardware (Arduino Due SAM3X8E) to avoid simulator discrepancies. Black-box (BB) tests drive inputs and observe outputs across module boundaries; white-box (WB) tests inspect internal state via Serial Monitor logs to verify inter-module data flow and HW/SW interface behaviour.

The 2oo2 dual-channel architecture is tested by operating both ECU boards simultaneously. The hardware AND gate enforces physical agreement — tests explicitly verify that a single-channel fault cannot fire any airbag LED.

Three categories of integration test are defined:

Category	Focus	Test Cases	Method
Normal Operation	Correct data flow and output under nominal sensor and communication conditions	IT-01 to IT-05	Black-Box (BB) — drive POTIs and switch; observe LEDs, Serial Monitor, USBCANV2
Fault Injection	Fault signals propagate across module boundaries and fail-safe outputs are asserted correctly	IT-06 to IT-10	BB and White-Box (WB) — inject faults via firmware or hardware; verify via Serial
Timing Verification	Real-time deadlines at module boundaries meet FSR-01, FSR-02, FSR-15, FSR-16, FSR-19	IT-11 to IT-15	Measurement — micros() timestamps via Serial; USBCANV2 frame timestamp analysis

3 Test Environment

3.1 Hardware Configuration

Component	Specification	Role
Arduino Due × 2	SAM3X8E ARM Cortex-M3, 84 MHz	ECU 1 (ID=1, TX=0x100) and ECU 2 (ID=2, TX=0x200)
POTI-1 (Accelerometer)	10 kΩ potentiometer on A0, 0–3.3 V → 0–100 g	Crash sensor simulation
POTI-2 (Gyroscope)	10 kΩ potentiometer on A1, 0–3.3 V → 0–360 °/s	Rollover sensor simulation
POTI-3 (Pressure)	10 kΩ potentiometer on A2, 0–3.3 V → 0–500 kPa	Side-impact sensor simulation
Baby Seat Switch	Momentary push switch on D11, 10 kΩ pull-up to 3.3 V, active-LOW	Occupant classification
DM74LS08N AND Gate	Quad 2-input AND, pins 1/2 → Airbag1 out, pins 4/5 → Airbag2 out	Hardware 2oo2 enforcement
Airbag1 LED (Green)	3 mm LED + 220 Ω from AND gate pin 3	Driver airbag deployment indicator
Airbag2 LED (Yellow)	3 mm LED + 220 Ω from AND gate pin 6	Passenger airbag deployment indicator
DWI LED (Red)	3 mm LED + 220 Ω shared between D10 of ECU1 and ECU2	Diagnostic Warning Indicator
Status LED (D13)	Built-in Arduino LED	Heartbeat / status blink
Waveshare SN65HVD230 × 2	CAN transceiver 3.3 V compatible	CAN bus interface for each ECU
USB-CAN-A adapter	500 kbit/s CAN to USB bridge	OMS PC CAN monitoring interface

3.2 Software / Tooling

Tool	Version	Role
Arduino IDE	2.3.6	Firmware compilation and upload
GCC ARM (avr-g++)	4.8.3 (bundled)	Cross-compiler for SAM3X8E
due_can library	v1.0 (open source)	CAN0 hardware abstraction
USBCANV2 (OMS PC)	Latest stable	Passive CAN frame monitoring at 500 kbit/s
Serial Monitor	Bundled in Arduino IDE	115200 baud diagnostic/Serial log capture
Git	Any	Firmware version control and release tagging

3.3 Network / Pin Connections

Both ECU boards share the same POTI-1/2/3 analog lines and the same baby seat switch (D11) as defined in HADS-AIRBAG-ECU-001 Rev 2.1. The shared CANH/CANL bus carries frames from both ECUs at 500 kbit/s. The USB-CAN-A adapter passively monitors this bus and must not inject frames during test execution unless specified by the test case.

4 Entry and Exit Criteria

4.1 Entry Criteria

The following conditions must be satisfied before test execution begins at any level:

#	Entry Criterion	Evidence Required
EC-01	Firmware build Rev corresponding to SADS-AIRBAG-ECU-001 Rev 2.0 complete and version-tagged in Git for both ECU 1 and ECU 2	Git tag present; confirmed in startup Serial banner
EC-02	HW/SW integration bring-up steps 1–12 (Section 6 of this plan) passed and signed off	Integration sign-off form with all 12 steps PASS
EC-03	Code review checklist (SADS Rev 2.0 Section 11 references ITP-AIRBAG-ECU-001) completed with all items PASS; two-person sign-off	Signed code review form
EC-04	Known gaps GAP-01 through GAP-08 documented with risk acceptance or closure plan	Gap status table in Section 9 of this plan completed
EC-05	Test hardware assembled per HADS-AIRBAG-ECU-001 Rev 2.1 wiring tables; continuity check passed; no short circuits	Continuity check sign-off by HW Engineer
EC-06	Serial Monitor confirms startup banner on both ECUs: document number, revision, ECU ID, build date/time	Captured Serial Monitor log from both ECUs at power-on
EC-07	USBCANv2 confirms CAN frames from ECU 1 (0x100) and ECU 2 (0x200) at approximately 500 ms intervals	USBCANv2 log showing dual-channel heartbeat prior to first test case

4.2 Exit Criteria

#	Exit Criterion
EX-01	All integration tests IT-01 through IT-15 executed; results recorded as PASS or FAIL with justification
EX-02	Statement coverage $\geq 100\%$ of executable statements demonstrated across integration test scope
EX-03	Branch coverage: all TRUE/FALSE outcomes of voteDeploy and localCrash conditionals exercised
EX-04	Fault detection coverage $\geq 90\%$ of identifiable single-point software faults detectable
EX-05	All FAIL results have an approved deviation report or corrective action plan
EX-06	Test evidence archived: Serial logs, USBCANv2 logs, LED photos/video, operator/date/FW version
EX-07	V&V lead and functional safety engineer sign-off obtained on integration test report

5 Integration Test Cases

The 15 integration test cases below are derived from SADS-AIRBAG-ECU-001 Rev 2.0 Section 9 (HW/SW Integration Procedure) and the module interfaces defined in Section 4.2. Tests are grouped into three categories: Normal Operation (IT-01 to IT-05), Fault Injection (IT-06 to IT-10), and Timing Verification (IT-11 to IT-15).

5.1 Normal Operation Tests

IT-01 — No-crash baseline — all sensors below threshold

Test ID	IT-01
Test Type	BB (Black-Box)
Test Name	No-crash baseline — all sensors below threshold
FSR Trace	FSR-04
ASIL	ASIL D
Preconditions	Full assembled system: both ECUs, CAN bus, AND gate, LEDs. USBCANV2 running. Serial Monitor open on both ECUs. All POTIs below threshold (ADC < 1000). Baby seat switch OPEN.
Test Procedure	Step 1: Confirm Serial on both ECUs shows localCrash=0, voteDeploy=0, systemFault=0. Step 2: Confirm all LEDs (Airbag1, Airbag2, DWI) are OFF. Step 3: Run continuously for a minimum of 5 minutes without touching any inputs. Step 4: Periodically verify Serial output every 60 seconds. Step 5: At end of run, photograph LED states and capture final Serial log.
Expected Result	Both Airbag LEDs and DWI LED remain OFF for the entire 5-minute run. Serial: localCrash=0, remoteCrash=0, voteDeploy=0 on both ECUs continuously.
Pass Criterion	Zero false deployments; Airbag LEDs OFF for ≥ 5 minutes without interruption.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

IT-02 — Full dual-channel crash — both ECUs above threshold simultaneously

Test ID	IT-02
Test Type	BB (Black-Box)
Test Name	Full dual-channel crash — both ECUs above threshold simultaneously
FSR Trace	FSR-01, FSR-03
ASIL	ASIL D
Preconditions	Full assembled system. Both ECUs communicating. USBCANV2 running. Serial Monitor open. Baseline: all POTIs below threshold, baby seat switch OPEN.
Test Procedure	Step 1: Confirm baseline — all LEDs OFF, Serial: voteDeploy=0. Step 2: Simultaneously advance POTI-1 above threshold on both ECUs (ADC > 1228, accel > 30.0 g). Step 3: Observe Airbag1 LED state immediately. Step 4: Verify Serial on ECU 1: localCrash=1, remoteCrash=1, voteDeploy=1. Step 5: Verify Serial on ECU 2: localCrash=1, remoteCrash=1, voteDeploy=1. Step 6: Check USBCANV2 — confirm bit 0 of byte 6 = 1 in both 0x100 and 0x200 frames. Step 7: Photograph LED states.
Expected Result	Airbag1 LED fires ON within one observed loop cycle. Both ECUs: voteDeploy=1 in Serial. USBCANV2: crash bit (byte 6, bit 0) = 1 in both CAN frames 0x100 and 0x200.
Pass Criterion	Airbag1 LED ON; Serial confirms voteDeploy=1 on both ECUs; USBCANV2 confirms crash bit in both CAN frames.

Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

IT-03 — Crash ceases — sensors returned below threshold

Test ID	IT-03
Test Type	BB (Black-Box)
Test Name	Crash ceases — sensors returned below threshold
FSR Trace	FSR-04
ASIL	ASIL D
Preconditions	System in crash state from IT-02 (Airbag1 LED ON, voteDeploy=1). Do not reset system between IT-02 and IT-03.
Test Procedure	Step 1: With system in crash state from IT-02, return POTI-1 below threshold on both ECUs (ADC < 1228). Step 2: Observe Airbag1 LED state immediately after POTI-1 drops below threshold. Step 3: Verify Serial on both ECUs: localCrash=0, voteDeploy=0 within one loop cycle. Step 4: Confirm system returns to ST-03 MONITORING state.
Expected Result	Airbag1 and Airbag2 LEDs extinguish within one loop cycle (~1 ms) of sensor drop below threshold. Serial: localCrash=0, voteDeploy=0 on both ECUs.
Pass Criterion	LEDs extinguish within one cycle of sensor drop below threshold.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

IT-04 — Baby seat suppression during crash

Test ID	IT-04
Test Type	BB (Black-Box)
Test Name	Baby seat suppression during crash
FSR Trace	FSR-09
ASIL	ASIL D
Preconditions	Full assembled system. Both ECUs in MONITORING. Baby seat switch OPEN (babySeat=FALSE).
Test Procedure	Step 1: Set both ECUs into crash state — advance POTI-1 above threshold on both ECUs. Step 2: Confirm Airbag1 LED ON and Airbag2 LED ON (no baby seat). Step 3: While crash state maintained, press and hold baby seat switch (D11 pulled LOW). Step 4: Observe Airbag1 and Airbag2 LED states after switch press. Step 5: Verify Serial on both ECUs: babySeat=1, D9=LOW. Step 6: Release switch. Verify Airbag2 LED re-activates within one cycle.
Expected Result	With baby seat switch pressed: Airbag1 LED ON (D8 HIGH), Airbag2 LED OFF (D9 LOW). On switch release: Airbag2 LED returns ON within one loop cycle.
Pass Criterion	Airbag2 LED stays OFF while baby seat switch pressed; Airbag1 LED remains ON during baby seat + crash condition.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	GAP-01: baby seat debounce (10 ms) not yet implemented. Switch bounce during crash vibration may cause intermittent results. Deviation accepted for lab scope.

IT-05 — CAN heartbeat visible on USBCANV2

Test ID	IT-05
Test Type	BB (Black-Box)
Test Name	CAN heartbeat visible on USBCANV2
FSR Trace	FSR-16
ASIL	ASIL B
Preconditions	Full assembled system. Both ECUs communicating. USBCANV2 running with timestamp logging enabled.
Test Procedure	Step 1: Allow system to run at baseline for 2 minutes without input changes. Step 2: Enable frame timestamp logging in USBCANV2. Step 3: Export USBCANV2 frame log after 2 minutes. Step 4: For ECU 1 (0x100) and ECU 2 (0x200), extract timestamps of 20 consecutive frames each. Step 5: Calculate inter-frame interval for each consecutive pair. Step 6: Record minimum, maximum, and mean interval for each ECU.
Expected Result	Frames for both 0x100 and 0x200 arrive at ~500 ms intervals. All inter-frame gaps within 450–550 ms.
Pass Criterion	All 19 intervals per ECU (from 20 frames) within 450–550 ms for both CAN IDs.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

5.2 Fault Injection Tests**IT-06 — Single-channel crash — only ECU1 above threshold, ECU2 below**

Test ID	IT-06
Test Type	BB (Black-Box)
Test Name	Single-channel crash — only ECU1 above threshold, ECU2 below
FSR Trace	FSR-03, FSR-08
ASIL	ASIL D
Preconditions	Both ECUs communicating. Baseline configuration. All LEDs OFF.
Test Procedure	Step 1: Confirm baseline — all LEDs OFF, voteDeploy=0 on both ECUs. Step 2: Advance POTI-1 above threshold on ECU 1 only (ADC > 1228). Keep all ECU 2 POTIs below threshold. Step 3: Observe Airbag1 LED. Verify LED remains OFF. Step 4: Check ECU 1 Serial: localCrash=1, remoteCrash=0, voteDeploy=0. Step 5: Check ECU 2 Serial: localCrash=0, remoteCrash=1 (received via CAN), voteDeploy=0. Step 6: Confirm AND gate output (Airbag1 LED) remains OFF. Step 7: Return POTI-1 below threshold. Confirm return to baseline.
Expected Result	Both Airbag LEDs OFF throughout. ECU 1: localCrash=1 but voteDeploy=0 (remoteCrash=0). ECU 2: remoteCrash=1 but voteDeploy=0 (localCrash=0). AND gate hardware prevents single-channel deployment.
Pass Criterion	No airbag LED fires when only one ECU detects a crash; AND gate hardware prevents single-channel deployment.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	This is the primary test of the 2oo2 hardware enforcement boundary. FAIL here represents a fundamental safety architecture failure.

IT-07 — CAN bus disconnected — peer timeout after 2000 ms

Test ID	IT-07
Test Type	BB (Black-Box)
Test Name	CAN bus disconnected — peer timeout after 2000 ms
FSR Trace	FSR-15, FSR-20
ASIL	ASIL D
Preconditions	Both ECUs communicating normally. Serial Monitor showing faultCOMM=0 on both ECUs. Stopwatch ready.
Test Procedure	Step 1: Record millis() from Serial output on both ECUs as reference. Step 2: Physically disconnect the CAN bus (remove CANH or CANL wire). Step 3: Start stopwatch from moment of disconnection. Step 4: Monitor Serial on both ECUs. Observe when faultCOMM transitions from 0 to 1. Step 5: Observe DWI LED — record time from disconnection to DWI LED ON. Step 6: Once DWI is ON, advance all POTIs above threshold. Verify both airbag LEDs remain OFF. Step 7: Calculate time from disconnection to faultCOMM=1 transition.
Expected Result	faultCOMM=TRUE on both ECUs within 2000 ms of CAN disconnection. DWI LED ON within 2000 ms. D8=LOW, D9=LOW. Both airbag LEDs remain OFF even with all sensors above threshold.
Pass Criterion	DWI ON within 2000 ms of CAN disconnection; airbag LEDs remain OFF when faultCOMM active.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	GAP-08: COMM_TIMEOUT = 2000 ms vs FSR-15 requirement of ≤50 ms. Safety rationale: CAN loss causes deployment inhibit (safe direction). Formal deviation required before production.

IT-08 — CAN reconnected after timeout — system recovers to MONITORING

Test ID	IT-08
Test Type	BB + WB
Test Name	CAN reconnected after timeout — system recovers to MONITORING
FSR Trace	FSR-15
ASIL	ASIL D
Preconditions	System in ST-04 FAULT_ACTIVE state following IT-07. Do not reset ECUs between IT-07 and IT-08.
Test Procedure	Step 1: With system in fault state from IT-07, reconnect CAN bus (reinsert CANH/CANL wire). Step 2: Monitor Serial on both ECUs. Observe when faultCOMM transitions from 1 to 0. Step 3: Observe DWI LED — confirm it extinguishes when faultCOMM clears. Step 4: Confirm Serial: systemFault=0 on both ECUs after faultCOMM clears. Step 5: Record millis() at faultCOMM=0. Calculate recovery time from CAN reconnection. Step 6: Confirm USBCANV2 shows frames from both 0x100 and 0x200 resuming.
Expected Result	faultCOMM clears to FALSE within one CAN heartbeat period (~500 ms) of CAN reconnection. DWI LED extinguishes. System transitions back to ST-03 MONITORING.
Pass Criterion	System recovery to MONITORING state within one CAN heartbeat period after reconnection; DWI extinguishes.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

IT-09 — Remote fault bit set — peer broadcasts fault in CAN byte 6

Test ID	IT-09
Test Type	WB (White-Box)
Test Name	Remote fault bit set — peer broadcasts fault in CAN byte 6
FSR Trace	FSR-03
ASIL	ASIL D
Preconditions	A fault-injection firmware build for ECU 2 that forces bit 2 of CAN byte 6 (systemFault bit) to 1 is required. ECU 1 running standard firmware.
Test Procedure	Step 1: Flash ECU 2 with fault-injection build (systemFault=TRUE in CAN status byte, bit 2 = 1). Step 2: Power up both ECUs. Verify USBCANV2 shows ECU 2 (0x200) frame with byte 6 bit 2 = 1. Step 3: Verify ECU 1 Serial: remoteFault=TRUE received. Step 4: Advance all POTIs above threshold on both ECUs simultaneously. Step 5: Observe Airbag1 and Airbag2 LEDs. Verify both remain OFF. Step 6: Verify ECU 1 Serial: localCrash=1, remoteCrash=1, remoteFault=TRUE, voteDeploy=0. Step 7: Restore ECU 2 standard firmware after test.
Expected Result	Both Airbag LEDs remain OFF. ECU 1: remoteFault=TRUE; voteDeploy=0 even with both sensors above threshold and remoteCrash=TRUE.
Pass Criterion	No airbag LED fires when remoteFault=TRUE, regardless of sensor state. voteDeploy=0 confirmed in Serial.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	Requires fault-injection firmware build for ECU 2. Restore standard firmware and re-verify IT-01 baseline before proceeding to IT-10.

IT-10 — POST failure simulation — faultCPU forced TRUE on ECU1

Test ID	IT-10
Test Type	BB (Black-Box)
Test Name	POST failure simulation — faultCPU forced TRUE on ECU1
FSR Trace	FSR-17, FSR-20
ASIL	ASIL D
Preconditions	Fault-injection firmware build for ECU 1 with cpuTest() hard-coded to return FALSE. ECU 2 running standard firmware.
Test Procedure	Step 1: Flash ECU 1 with POST-fail firmware build. Step 2: Power-cycle both ECUs simultaneously. Step 3: Observe ECU 1 Serial: confirm faultCPU=1, systemFault=1, DWI LED ON from first cycle. Step 4: Advance all POTIs above threshold on both ECUs. Attempt to trigger crash state. Step 5: Observe Airbag1 LED throughout entire session (minimum 60 seconds). Step 6: Verify Airbag1 LED remains OFF for entire session. Step 7: Restore standard ECU 1 firmware after test.
Expected Result	DWI LED ON from ECU 1 startup (first loop iteration). Airbag1 LED remains OFF for entire 60-second session. POST fault flags are sticky — not cleared by any sensor state.
Pass Criterion	No airbag LED fires throughout entire session when ECU 1 has POST fault. DWI ON from startup.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	Always restore standard firmware after test and re-execute bring-up steps 3 and 5 before proceeding.

5.3 Timing Verification Tests

IT-11 — Loop cycle time ≤ 1 ms

Test ID	IT-11
Test Type	Timing Measurement
Test Name	Loop cycle time ≤ 1 ms
FSR Trace	FSR-02
ASIL	ASIL D
Preconditions	Timing-instrumented firmware build with micros() logged at loop() start and end via Serial. System in MONITORING state.
Test Procedure	Step 1: Flash timing-instrumented firmware to both ECUs. Step 2: Open Serial Monitor at 115200 baud. Enable log capture to file. Step 3: Run for at least 10,000 complete loop iterations. Allow Serial log to accumulate. Step 4: Export Serial log. Parse cycle time values (end_micros – start_micros) for each iteration. Step 5: Calculate: maximum, mean, and 99th percentile cycle time. Step 6: Repeat with all three POTIs above threshold (crash state) to measure worst-case timing. Result fields — Normal: Max=___ μ s Mean=___ μ s 99th %ile=___ μ s PASS/FAIL:___ Result fields — Crash: Max=___ μ s Mean=___ μ s 99th %ile=___ μ s PASS/FAIL:___
Expected Result	99th percentile cycle time ≤ 1000 μ s (1 ms) in both normal and crash state.
Pass Criterion	99th percentile cycle time ≤ 1000 μ s measured over 10,000 iterations in both operating modes.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	Serial.print() must be gated behind the 1000 ms elapsed check during this test to avoid print overhead inflating cycle time.

IT-12 — Crash detection response ≤ 20 ms

Test ID	IT-12
Test Type	Timing Measurement
Test Name	Crash detection response ≤ 20 ms
FSR Trace	FSR-01
ASIL	ASIL D
Preconditions	Timing-instrumented firmware build. Serial Monitor with log capture active. POTI-1 set to just below threshold (ADC ~ 1220).
Test Procedure	Step 1: Set POTI-1 to just below threshold (ADC ~ 1220 , accel ~ 29.8 g). Step 2: Note millis()/micros() at the cycle immediately before POTI-1 is advanced. Step 3: Quickly advance POTI-1 above threshold (ADC > 1228). Step 4: Note millis()/micros() at the first cycle where localCrash=1 appears in Serial. Step 5: Calculate delta = (timestamp at localCrash=1) – (timestamp at last localCrash=0). Step 6: Repeat steps 2–5 for 10 independent trials. Trials: 1:___ ms 2:___ ms 3:___ ms 4:___ ms 5:___ ms 6:___ ms 7:___ ms 8:___ ms 9:___ ms 10:___ ms Max:___ ms
Expected Result	All 10 delta values ≤ 20 ms. Detection typically occurs within 1–2 ms given ≤ 1 ms cycle time.
Pass Criterion	All 10 trial deltas ≤ 20 ms.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

IT-13 — CAN heartbeat interval 500 ms ± 50 ms

Test ID	IT-13
Test Type	Timing Measurement
Test Name	CAN heartbeat interval 500 ms ± 50 ms
FSR Trace	FSR-16
ASIL	ASIL B
Preconditions	Both ECUs in MONITORING state. USBCANV2 running with hardware timestamp logging enabled.
Test Procedure	Step 1: Allow system to run at baseline for 2 minutes. Step 2: Export USBCANV2 timestamp log. Step 3: ECU 1 (0x100): extract timestamps of 20 consecutive frames; calculate 19 inter-frame intervals. Step 4: ECU 2 (0x200): repeat step 3. Step 5: Record minimum, maximum, and mean interval for each ECU. ECU 1: Min=___ ms Max=___ ms Mean=___ ms All within 450–550 ms? PASS/FAIL:___ ECU 2: Min=___ ms Max=___ ms Mean=___ ms All within 450–550 ms? PASS/FAIL:___
Expected Result	All 19 inter-frame intervals for both 0x100 and 0x200 within 450 ms to 550 ms.
Pass Criterion	All 38 intervals (19 per ECU) within 450–550 ms.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	

IT-14 — faultCOMM declared ≤ 2000 ms after CAN loss

Test ID	IT-14
Test Type	Timing Measurement
Test Name	faultCOMM declared ≤ 2000 ms after CAN loss
FSR Trace	FSR-15
ASIL	ASIL B
Preconditions	Both ECUs communicating. Serial Monitor with log capture active on both ECUs.
Test Procedure	Step 1: Note millis() of the last received CAN heartbeat from Serial log (lastRemoteHB update). Step 2: Physically disconnect CAN bus. Step 3: Monitor Serial. Record millis() at exact moment faultCOMM transitions from 0 to 1. Step 4: Calculate: delta = millis(faultCOMM=1) – millis(last valid CAN frame). Step 5: Repeat 3 times. Trial 1: Last HB=___ faultCOMM=___ Delta=___ ms PASS/FAIL:___ Trial 2: Last HB=___ faultCOMM=___ Delta=___ ms PASS/FAIL:___ Trial 3: Last HB=___ faultCOMM=___ Delta=___ ms PASS/FAIL:___
Expected Result	faultCOMM declared within 2050 ms of last valid CAN frame on all 3 trials.
Pass Criterion	All three measured deltas ≤ 2050 ms. NOTE: GAP-08 — FSR-15 requires ≤50 ms; current implementation uses 2000 ms; deviation documented.
Evidence Required	Timestamped Serial log from both ECUs. USBCANV2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	GAP-08: COMM_TIMEOUT = 2000 ms deviates from FSR-15 requirement of ≤50 ms. Formal deviation required before production.

IT-15 — DWI activates within 800 ms of fault

Test ID	IT-15
Test Type	Timing Measurement
Test Name	DWI activates within 800 ms of fault

FSR Trace	FSR-19
ASIL	ASIL A
Preconditions	Both ECUs in MONITORING state. Serial Monitor with log capture active.
Test Procedure	Step 1: Record millis() as reference. Step 2: Inject fault by disconnecting CAN bus (will trigger faultCOMM → systemFault after COMM_TIMEOUT), OR use POST-fail firmware build on ECU 1 (DWI activates at first loop iteration). Step 3: Record millis() at fault injection (CAN disconnect or power-on). Step 4: Record millis() at first cycle where D10=HIGH in Serial or observe DWI LED ON. Step 5: Calculate delta = millis(D10=HIGH) – millis(fault_inject). Step 6: Repeat 5 times. Trials: 1:___ ms 2:___ ms 3:___ ms 4:___ ms 5:___ ms Max:___ ms
Expected Result	D10 transitions HIGH (DWI LED ON) within ≤800 ms of fault onset on all 5 trials.
Pass Criterion	All 5 measured deltas ≤ 800 ms.
Evidence Required	Timestamped Serial log from both ECUs. USBCANv2 CAN frame log. LED state photograph or video. Operator name, date, firmware version (from startup Serial banner).
Result (Pass/Fail)	
Notes	POST-fail method provides direct measurement independent of COMM_TIMEOUT; DWI activates within ~1 ms. CAN disconnect method activates DWI at ~2000 ms due to GAP-08.

6 HW/SW Integration Bring-Up Procedure

The following 12-step procedure is derived from SADS-AIRBAG-ECU-001 Rev 2.0 Section 9.1. It must be completed and signed off before integration test cases IT-01 through IT-15 are executed. Execution results and sign-off evidence are recorded in TR-AIRBAG-ECU-001.

Step	Action	Acceptance Criterion	Owner	Pass/Fail
1	Assemble hardware per HADS-AIRBAG-ECU-001 Rev 2.1 — breadboard wiring, potentiometers, baby seat switch, CAN transceivers, AND gate, LEDs.	All wiring verified against HADS Tables 5–14. No short circuits. Continuity check on all signal lines.	HW Engineer	
2	Verify power rails: 5 V to AND gate VCC (pin 14); 3.3 V to CAN transceivers (Waveshare VCC); GND common to all.	Measured: 5 V ± 0.25 V at AND gate pin 14. 3.3 V ± 0.15 V at Waveshare VCC. GND continuity confirmed.	HW Engineer	
3	Flash ECU 1 firmware (ECU_ID=1, DATA_ID=0x100, REMOTE_ID=0x200) via Arduino IDE 2.3.6.	Upload completes 'Done uploading'. Serial Monitor (115200 baud) shows startup banner with ECU ID: 1 and build date/time.	SW Engineer	
4	Flash ECU 2 firmware (ECU_ID=2, DATA_ID=0x200, REMOTE_ID=0x100) via Arduino IDE 2.3.6.	Upload completes. Serial Monitor shows startup banner with ECU ID: 2.	SW Engineer	
5	Verify POST results on both ECUs via Serial Monitor immediately after power-on.	Serial: faultCPU=0, faultRAM=0, faultADC=0 on both ECUs. D13 (status LED) blinking at ~500 ms.	SW Engineer	
6	Connect OMS PC via USB-CAN-A adapter. Open USBCANv2 at 500 kbit/s.	CAN frames with ID 0x100 and 0x200 visible at ~500 ms intervals in USBCANv2 log within 5 seconds of connection.	HW/SW Engineer	

Step	Action	Acceptance Criterion	Owner	Pass/Fail
7	Verify CAN inter-ECU communication — check faultCOMM=0 on both ECUs via Serial Monitor.	Serial: faultCOMM=0 on both ECUs. remoteCrash and remoteFault bits correctly parsed (both FALSE at baseline).	SW Engineer	
8	Verify all output pins at rest (all sensors below threshold, no faults).	D8=LOW, D9=LOW, D10=LOW on both ECUs. D13 blinking. Airbag1 LED OFF, Airbag2 LED OFF, DWI LED OFF.	HW Engineer	
9	Verify AND gate logic — single-channel test: advance POTI-1 above threshold on ECU1 only; ECU2 below threshold.	Airbag1 LED remains OFF. AND gate output (pin 3) LOW. ECU1 Serial: localCrash=1, voteDeploy=0 (remoteCrash=0 from ECU2).	HW Engineer	
10	Verify full dual-channel crash — advance POTI-1 above threshold on both ECUs simultaneously.	Airbag1 LED ON (AND gate pin 3 HIGH). Both ECUs Serial: voteDeploy=1.	HW/SW Engineer	
11	Verify baby seat suppression — press baby seat switch during step 10 crash.	Airbag1 LED remains ON; Airbag2 LED OFF. D9 LOW on both ECUs.	HW/SW Engineer	
12	Sign off HW/SW integration — record results and release to functional testing. Sign-off recorded in TR-AIRBAG-ECU-001.	All steps PASS. Integration sign-off form completed.	Project Lead	

7 Coverage Targets

ISO 26262-6 ASIL D requires MC/DC (Modified Condition/Decision Coverage) or equivalent structural coverage. The agreed coverage targets for this laboratory implementation are:

Coverage Type	Target	Method	Test Cases
Statement Coverage	100% of executable statements	Verify all inter-module paths and integration-level statements exercised by at least one integration test case	IT-01 to IT-15
Branch Coverage	All TRUE and FALSE outcomes of every conditional	Explicit test cases exercising each TRUE/FALSE branch of voteDeploy, localCrash, systemFault, babySeat conditionals	IT-01 to IT-10
Fault Detection Coverage	≥ 90% of identifiable single-point software faults detectable	Fault injection tests cover CPU fault, RAM fault, ADC fault, CAN timeout fault, remote fault, baby seat stuck conditions	IT-06, IT-07, IT-09, IT-10

7.1 Conditional Branches Requiring Explicit Coverage

Conditional Expression	TRUE Case Covered By	FALSE Case Covered By
accel > THRESH_ACCEL (30.0 g)	IT-02 (POTI-1 above threshold)	IT-01 (all sensors below threshold)
gyro > THRESH_GYRO (120.0 °/s)	IT-02 (POTI-2 above threshold)	IT-01
press > THRESH_PRESS (200.0 kPa)	IT-02 (POTI-3 above threshold)	IT-01
localCrash (OR fusion)	IT-02 / IT-06 (single-sensor above threshold)	IT-01 (none above)
localCrash && remoteCrash	IT-02 (both ECUs crash)	IT-06 (single ECU only)
!systemFault	IT-02 (no fault)	IT-10 (POST fault forces systemFault)
!remoteFault	IT-02 (no fault)	IT-09 (peer broadcasts fault)
voteDeploy && !babySeat → D9	IT-02 (no baby seat during dual-channel crash)	IT-04 (baby seat switch active during crash)
millis() - lastRemoteHB > 2000	IT-07 (CAN disconnected > 2 s)	IT-05 (heartbeat normal)

8 Test Evidence and Reporting

8.1 Required Evidence per Test Case

The following evidence must be collected and archived for every test case executed:

Evidence Item	Format	Storage Location
Serial Monitor log (both ECUs)	Timestamped .txt file (115200 baud capture)	Test archive — date-stamped folder per test session
USBCANV2 CAN frame log	Exported .csv or .log from USBCANV2	Test archive — same folder as Serial logs
LED state photograph or video	.jpg / .mp4 capturing Airbag1, Airbag2, DWI LED states at key moments	Test archive — labelled with test case ID
Firmware version record	From startup Serial banner: document number, revision, ECU ID, build date/time (__DATE__, __TIME__)	Included in Serial log and test report header

Evidence Item	Format	Storage Location
Test operator record	Operator name, date, hardware revision, pass/fail determination	Test report sign-off sheet

8.2 Test Report Structure

A test report (TR-AIRBAG-ECU-001) shall be produced upon completion of all 15 integration test cases. The report shall include:

- Report header: document number, revision, date, firmware version tested, hardware revision, test environment configuration
- Summary table of all 15 test cases: Test ID, name, result (Pass/Fail), and any deviations
- For each FAIL: defect description, root cause (if known), corrective action plan or deviation approval reference
- Coverage summary against targets in Section 7
- Test evidence log listing all archived files with their paths and checksums
- Sign-off table completed by test executor, independent reviewer, V&V lead, and functional safety engineer

9 Compliance Gaps Affecting Test Scope

The following gaps between the current firmware and FSR requirements are identified in SADS-AIRBAG-ECU-001 Rev 2.0 Section 10. Each gap is assigned a handling status for this plan.

Gap ID	FSR	Description	Risk	Priority	Test Plan Status
GAP-01	FSR-11	Baby seat debounce not implemented — D11 read directly without 10 ms stability filter	Medium	High	IT-04 executes as-is. Switch bounce during crash vibration may cause intermittent false suppression. Deviation accepted for lab scope; closure required before production.
GAP-02	FSR-12	Baby seat line continuity monitoring not implemented — open-circuit stuck-HIGH not detected	High	High	No test case can verify unimplemented function. RR-04 residual risk accepted for lab scope. Closure required before production. IT-04 proceeds with functional switch only.
GAP-03	FSR-13	CRC not included in CAN frame — bytes 6–7 carry status and ECU_ID, not a 16-bit CRC	Medium	High	IT-05 and IT-13 verify frame content and timing without CRC verification. USBCANV2 log used to confirm byte content. CRC closure required before SG-07 can be fully validated.
GAP-04	FSR-14	Parallel CAN bitstream readback not implemented	Low	Medium	No integration test defined for unimplemented feature. Deferred. Hardware AND gate provides second-layer verification.
GAP-05	FSR-17	CAN communication not verified during POST — Can0.begin() return value not checked	Medium	Medium	POST integration test IT-10 proceeds without CAN init check. CAN functional verification provided by IT-05 / IT-06.
GAP-06	FSR-22	Hardware WDT not enabled in firmware	High	Critical	BLOCKING for production sign-off. All timing tests (IT-11 to IT-15) proceed. WDT implementation and closure required before validation sign-off. See SADS Rev 2.0 Section 8.1.
GAP-07	FSR-21	Fault logging to DTC/non-volatile memory not implemented	Low	Low	No integration test for EEPROM/flash DTC storage. Deferred. Serial Monitor log used as fault evidence substitute for lab scope.
GAP-08	FSR-15	CAN timeout threshold is 2000 ms vs FSR-15 requirement of ≤50 ms	Medium	High	IT-14 tests at 2000 ms threshold. Pass criterion set to ≤2050 ms reflecting implementation. Safety rationale: CAN loss → inhibit (safe direction). Formal deviation justification required before production.

10 Full Requirement Traceability Matrix

This matrix provides bidirectional traceability from each FSR requirement through implementing SWRs, implementing modules, and integration test cases. Open gaps are flagged. Derived from SADS-AIRBAG-ECU-001 Rev 2.0 Section 12.

FSR ID	FSR Title	ASIL	SWR(s)	Test Case(s)	Gap
FSR-01	Crash Detection Timing (≤20 ms)	ASIL D	SWR-04, SWR-05, SWR-06	IT-02, IT-06, IT-12	—
FSR-02	Algorithm Cycle Time (≤1 ms)	ASIL D	SWR-11	IT-11	—
FSR-03	Deployment Authorization Logic	ASIL D	SWR-10, SWR-12	IT-02, IT-04, IT-06, IT-09	—
FSR-04	Prevention of Unintended Deployment	ASIL D	SWR-06	IT-01, IT-03, IT-06	—
FSR-07	Dual-Channel Redundancy	ASIL D	SWR-17, SWR-18, SWR-19	IT-06	—
FSR-08	Channel Agreement Verification	ASIL D	SWR-12	IT-06	—
FSR-09	Passenger Airbag Suppression	ASIL D	SWR-07, SWR-13	IT-04	—
FSR-10	Suppress-on-Fault Strategy	ASIL D	SWR-09	IT-04	GAP-02
FSR-11	Digital Input Debounce (10 ms)	ASIL D	SWR-08	— (not yet implemented)	GAP-01
FSR-12	Baby Seat Line Monitoring	ASIL D	— (not yet implemented)	— (not yet implemented)	GAP-02
FSR-13	CRC-Protected CAN	ASIL B	SWR-20, SWR-21	IT-05, IT-13 (no CRC)	GAP-03
FSR-14	CAN Bitstream Readback	ASIL B	— (not yet implemented)	—	GAP-04
FSR-15	CAN Timeout Detection (≤50 ms)	ASIL B	SWR-18	IT-07, IT-14 (at 2000 ms)	GAP-08
FSR-16	Crash Notification Transmission	ASIL B	SWR-22	IT-05, IT-13	—
FSR-17	Power-On Self-Test	ASIL C	SWR-01, SWR-02, SWR-03	IT-10	GAP-05
FSR-18	Continuous Diagnostic Monitoring	ASIL C	SWR-23, SWR-24	IT-07	—
FSR-19	DWI Activation (≤800 ms)	ASIL A	SWR-23, SWR-24	IT-15	—
FSR-20	Safe-State Transition	ASIL D	SWR-14	IT-07, IT-10	—
FSR-21	Fault Logging (DTC)	ASIL A	— (not yet implemented)	—	GAP-07
FSR-22	Memory Integrity / WDT	ASIL D	SWR-15, SWR-16	IT-10 (partial)	GAP-06

11 Approval and Sign-Off

Role	Name	Signature	Date
Software Safety Engineer			
Hardware Safety Engineer			
Functional Safety Engineer			
V&V Lead			
Project Supervisor			